



The New Safe Confinement (Chernobyl Arch)

By Conner Glover



The Unseen Threat: Chernobyl's Legacy

01

April 26, 1986

Reactor No. 4 at Chernobyl Nuclear Power Plant suffers catastrophic explosion, releasing massive amounts of radioactive material into the atmosphere

02

Hasty Containment

The original "Object Shelter" (OS) built within months to contain radioactive materials—never intended as permanent solution

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Decades of Danger

Years of structural decay and constant radiation threat posed risk of catastrophic dissemination across Europe

A Monumental Challenge: The Need for a New Barrier



Purpose & Vision

Create permanent containment for radioactive substances while enabling safe deconstruction of the deteriorating old shelter



International Effort

Funded by Chernobyl Shelter Fund (CSF) managed by European Bank for Reconstruction and Development, involving contributions from 45 nations



Extreme Engineering

One-of-a-kind structure designed specifically for extreme radiation conditions and long-term environmental protection

Engineering Marvel: The Arch Design

257m

Span Width

Wide enough to cover the Statue of Liberty laid on its side

162m

Length

Extends beyond the original shelter footprint

108m

Height

Taller than London's Big Ben clock tower

36K

Weight (Tons)

Final skidded weight equivalent to 4,500 elephants

Designed for 100-year operational lifespan, resistant to seismic impacts up to magnitude 6 and Tornado Class 3 winds



The NSC: A Shield Against Radiation

A vast, gleaming steel arch protecting millions from radioactive threat



Construction Under Extreme Conditions



NOVARKA Joint Venture

Engineering and construction performed by NOVARKA, a collaboration between VINCI and BOUYGUES



Local Partnership

Approximately 50 Ukrainian organizations participated, creating jobs while ensuring local expertise and knowledge transfer



Remote Assembly

Built in sections adjacent to the old shelter, then skidded into final position to minimize worker radiation exposure

Ensuring Containment: Leak-Tightness and Safety

Annular Space System

A carefully engineered gap between internal and external claddings maintains positive ventilation pressure, preventing radioactive particles from escaping

Elastic Membrane Seal

Flexible connection between arch and existing structures ensures leak-tightness while accommodating structural movement and minimizing impact on aging foundations

Complete Envelope

Designed to fully envelope the old Object Shelter and adjacent contaminated territory, creating comprehensive containment zone



Beyond the Arch: Auxiliary Facilities and Future Operations



Technological Building

Central hub for monitoring systems and operational control



Electrical Equipment Building

Power distribution and backup systems ensuring continuous operation



Fire Water Pump Station

Fire suppression infrastructure protecting the facility



Diesel Power Stations

Emergency power supply maintaining safety systems during grid failures



Storm Water Treatment

Environmental protection preventing contaminated runoff



Overhead Cranes

Two main cranes suspended within arch facilitate dismantling old shelter

A Symbol of Resilience and Progress

International Cooperation

A testament to what humanity can achieve when nations unite for common purpose, 45 countries contributed to this engineering marvel

Environmental Protection

A critical step in securing the Chernobyl site and mitigating long-term environmental risks for generations to come

Century of Safety

Built to last 100 years, providing time and conditions for future deconstruction operations and creating ecologically safe system

"The New Safe Confinement stands as a beacon of hope—a monument to human ingenuity, perseverance, and our commitment to protecting future generations from nuclear disaster."

